

LECTURE SERIES · ISEA

Intelligent Digital Forensics: LLMs, NLP, and Explainable AI for Cyber Investigations

Introduction to Intelligent Digital Forensics and Cyber Investigations

Day 1 · Foundations

Session 1

2 Hours

Lecture + Case Study

Digital evidence · investigative workflows · the role of AI in modern cyber investigations

Ten Sessions, One Arc

DAY 1 Foundations

DAY 2 LLMs & AI Threats

DAY 3 Explainable & Trustworthy AI

DAY 4 Social Media & Threat Intel

DAY 5 Emerging Directions

S1 YOU ARE HERE

Intro to Intelligent Digital Forensics

S3

LLMs in Cyber Investigations

S5

Explainable AI (XAI)

S7

Social Media Forensics

S9

Emerging Trends

S2

NLP for Cybersecurity & Forensics

S4

Misinformation & Deepfakes

S6

Ethics & Responsible AI

S8

Cyber Threat Intelligence

S10

Future Directions & Research

Every day: lecture sessions + a case study (20–30 min) + a guided hands-on activity (30–40 min) — all on free, cloud-based tools. No GPUs, no special hardware.

By the End of This Session, You Can...

- 1 **Trace** how digital forensics evolved from disk recovery to AI-assisted, intelligence-driven investigation.
- 2 **Apply** the core principles of a sound cyber investigation — legality, chain of custody, integrity, repeatability.
- 3 **Classify** digital evidence types and map them onto the six-stage forensic workflow.
- 4 **Explain** where AI genuinely assists cyber defence and forensics — and where it must not replace human judgement.
- 5 **Recognise** language as forensic evidence, and identify red flags in a real suspicious email.
- 6 **Anticipate** the challenges AI-generated content creates for attribution, verification, and admissibility.

Objectives follow Bloom's ladder — from remembering the history to analysing real evidence by the end.

How the Next 120 Minutes Run



	Core lecture	Topics 1–7 with cases and short discussions	90 min
	Case study	Investigating suspicious email communications	20 min
	Q&A	Open discussion	10 min

Seven Topics, One Case, One Exercise

1

Evolution of Digital Forensics

Four eras — and why 'intelligent' forensics emerged

2

Fundamentals of Cyber Investigations

Principles, actors, and the legal foundation in India

3

Digital Evidence & Workflows

Evidence types, metadata, and the six-stage process

4

Role of AI in Cybersecurity

What machine intelligence actually does — and its limits

5

Language-Based Evidence

Text, style, and headers as investigative signal

6

Challenges in AI-Driven Threats

The adversary has AI too: deepfakes, GenAI phishing

7

Emerging Trends

LLMs, XAI, and the road through Sessions 2–10



Case Study

Suspicious email investigation · evidence identification exercise

How confident are you in digital forensics?

📱 Join the Wooclap session by scanning the QR code or entering the event code.

Go to **wooclap.com**

Enter the event code in the top banner

Event code

XMTLMHU



TOPIC 1 OF 7

Evolution of Digital Forensics in Modern Cyber Ecosystems

01

- From seized floppy disks to evidence scattered across clouds and platforms
- Why data volume forced the discipline to become 'intelligent'
- A famous case where a single file's metadata closed a 30-year hunt

What Is Digital Forensics?

The scientific identification, preservation, examination, analysis, and presentation of digital evidence in a manner that is legally acceptable — i.e., that survives challenge in court.



It is a science

Method, documentation, repeatability — not intuition.
Findings must be reproducible by another examiner.



It is legal-facing

Every technical step is shaped by what a court or tribunal will later accept as reliable.

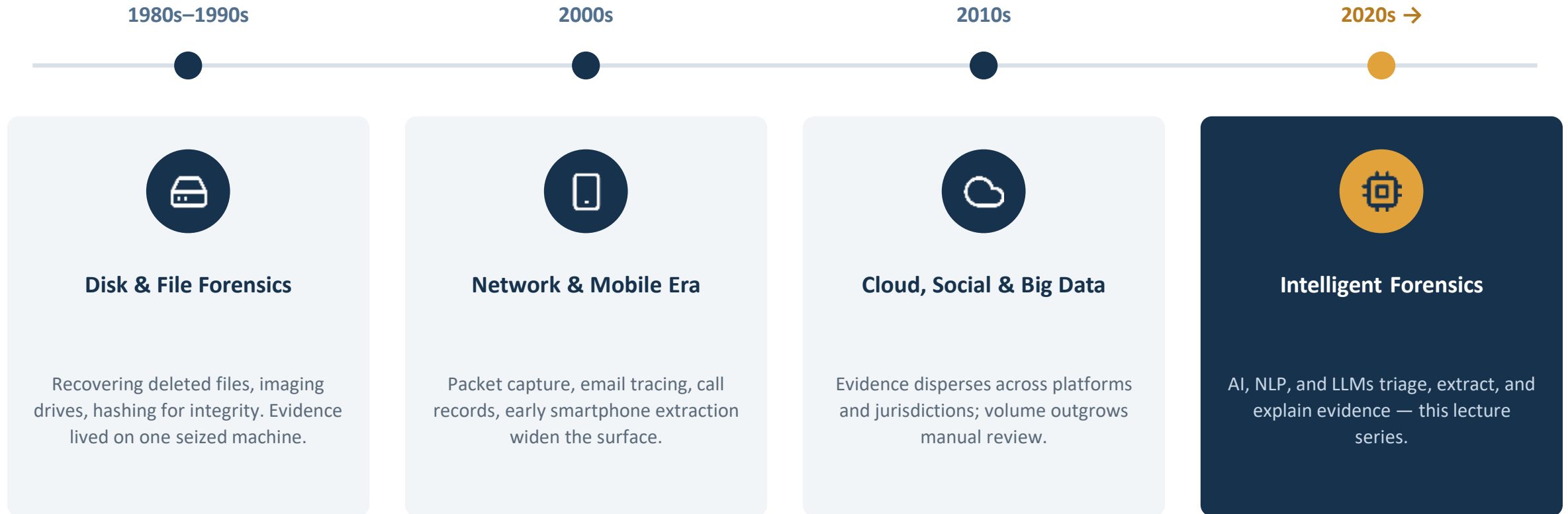


It is Locard, digitised

“Every contact leaves a trace.” Online, every action leaves logs, metadata, and artefacts.

Forensics ≠ cybersecurity: security tries to prevent the incident; forensics reconstructs it afterwards.

Four Eras of Digital Forensics



Each era did not replace the last — it stacked on top. Today's examiner needs all four skill sets.

What Actually Changed

CLASSICAL FORENSICS (THEN)

INTELLIGENT FORENSICS (NOW)



One device, physically seized



Hundreds of sources across cloud, apps, and platforms



Recover artefacts: files, fragments



Interpret behaviour: patterns, language, coordination



Weeks of manual examination



Machine triage in minutes, expert review of what matters



Examiner reads everything



Examiner supervises models that read everything



Testify about what was found



Also testify about how the AI found it — explainability

Where Evidence Lives Today



Cloud services

Storage, SaaS logs, collaboration platforms



Mobile devices

Messaging apps, location traces, app artefacts



Social platforms

Posts, DMs, account metadata, engagement trails



IoT & smart devices

Sensor logs, connection histories, telemetry



Encrypted channels

E2E messengers — metadata often outlives content



Enterprise systems

SIEM alerts, access logs, email archives

VOLUME

far more data than analysts can read

VELOCITY

evidence appears, spreads, vanishes fast

VARIETY

text, media, logs, metadata mixed together

Where Would You Look First?

📱 Join the Wooclap session by scanning the QR code or entering the event code.

Go to **wooclap.com**

Enter the event code in the top banner

Event code

XMTLMHU



The Scale Problem, In Numbers



256+ GB

of data on a single modern smartphone —
every byte potentially evidentiary in a seizure



≈4.8 million

estimated global shortfall of cybersecurity
professionals (ISC² 2024 workforce study)



101,928

cybercrime cases registered in India in 2024 —
NCRB, Crime in India report

Forensic backlogs are not a staffing failure — they are an arithmetic failure. Machines must read first.

BTK: One File How Metadata Solved a 30-Year Mystery

1974–1991

The self-named 'BTK' (Bind, Torture, Kill) killer murders ten people in Wichita, Kansas — and taunts police with letters.

2004

After years of silence, he resumes contact with media and police, asking if a floppy disk could be traced.

Feb 2005

Police say no (a lie). He mails a floppy disk with a Word document to a TV station.

The find

Deleted-file metadata shows 'Christ Lutheran Church' and the author name 'Dennis'.

The arrest

Dennis Rader — church council president — is arrested within days. He pleads guilty.



Why it matters here

Evidence the suspect never knew existed — embedded metadata — did what three decades of conventional policing could not.

Intelligent forensics generalises this: systematically surfacing the traces humans overlook, at scale.

What metadata do the documents you create every day carry?

TOPIC 2 OF 7

Fundamentals of Cyber Investigations

02

- Types of investigations and who conducts them in India
- Five principles every credible investigation obeys
- Chain of custody, hashing, and the legal foundation: IT Act & Section 65B

Four Kinds of Cyber Investigation



Criminal

Police / central agencies investigating offences: fraud, hacking, CSAM, cyber terrorism. Highest evidentiary standards — proof beyond reasonable doubt.



Civil

Disputes between parties: IP theft, defamation, contract breaches. Standard: balance of probabilities. E-discovery is the workhorse.



Corporate / internal

Employers probing policy violations, insider leaks, harassment. Constrained by employment law and privacy obligations.



Incident response

Live investigation during an ongoing attack. Tension: restore services fast vs preserve evidence intact. Both matter.

Same techniques, different rules: who may collect what, and what standard the findings must meet.

Who Investigates Cybercrime in India



CERT-In (MeitY)

National nodal agency for cyber incident response. Its 2022 Directions require reporting of specified incidents within six hours.



I4C (MHA)

Indian Cyber Crime Coordination Centre — runs the 1930 helpline and the cybercrime.gov.in national reporting portal.



State cyber cells

Police cyber crime units register and investigate cases; specialised cyber police stations in most states.



Forensic labs

CFSL / State FSLs conduct examinations; their reports carry evidentiary weight under Indian law.



NCRB

Compiles national crime statistics and publishes the annual Crime in India report. The latest report shows continued year-on-year growth in cybercrime, highlighting the increasing demand for specialised cyber investigations.



Sectoral CERTs & RBI

Finance, power, and defence maintain sector response teams; RBI mandates bank incident reporting.

Practical detail participants ask about: victims dial 1930 or file at cybercrime.gov.in — speed matters for freezing fraudulent transfers.

Five Rules of a Sound Investigation



Legality & authorisation

Act within jurisdiction and mandate — search authority, warrants, and platform terms decide what evidence is usable.



Chain of custody

Every hand-off documented. If custody breaks, even genuine evidence can be challenged and excluded.



Integrity & minimal alteration

Work on verified copies; hash originals so any change is detectable.



Repeatability

Another qualified examiner following your notes should reach the same result with the same data.



Contemporaneous documentation

Record tools, versions, commands, timestamps as you go — never reconstructed afterwards.

These five survive every technology shift — including AI. Automation never replaces accountability.

Chain of Custody, Concretely

A custody record answers, for every moment of the evidence's life:



What

Exact item + unique ID: “Samsung phone, IMEI ..., Exhibit P-7” — plus hash values of any images taken.



Who

Every person who touched it, with signatures — seizing officer, transporter, examiner, storage custodian.



When & where

Timestamped locations for seizure, transport, storage, and every access — gaps are what defence counsel hunts for.



Why & how

Purpose of each access and what was done — imaged, examined, re-sealed — with tools and versions named.

Courtroom reality: the cheapest way to defeat digital evidence is not to dispute the bytes — it is to show the custody record has a hole.

Arjun Panditrao Khotkar v. Kailash Kushanrao Gorantyal (2020)

An election dispute relied on **electronic evidence (video recordings)**. The Supreme Court had to decide whether such evidence could be admitted **without a Section 65B certificate**.

Judgment: The Court ruled that a **Section 65B certificate is mandatory** for secondary electronic evidence (now **Section 63 of the Bharatiya Sakshya Adhiniyam, 2023**).

Key Takeaway:

Genuine digital evidence may still be rejected by a court if it is not accompanied by the required legal certificate.

This case established that **legal compliance is as important as technical accuracy** in digital forensics.

Hashing: The Tamper-Evident Seal

A cryptographic hash (MD5, SHA-256) is a fixed-length fingerprint of data. Change one character and the fingerprint changes completely — the 'avalanche effect'.

Original: “Transfer Rs 48,00,000 today”

SHA-256: d09b717b21d7995278e0dc6a1334dc5e571138009f5e2faf1c28a12cefb145a7

One digit changed: “Transfer Rs 48,00,001 today”

SHA-256: 8705495001600eebe08297907da6f591fd288fd40fca7a24a24f7ab74c46eebd



At acquisition

Hash the original, image it, hash the image — matching values prove a faithful copy.



At every stage after

Re-hash before analysis and before court: identical value = untouched evidence.

These two hashes above are real SHA-256 values — computed for this slide, not illustrative.

The Law That Frames All of This (India)



IT Act, 2000 (amended 2008)

Information Technology Act, 2000 (amended 2008) remains India's principal legislation governing cyber offences and electronic records. Key provisions include Sections 43, 66, 66C, 66D, and 67, covering unauthorised access, computer-related offences, identity theft, cheating by personation, and obscene electronic content.



Evidence Act s.65B → BSA 2023 s.63

Section 65B of the Indian Evidence Act has been replaced by Section 63 of the Bharatiya Sakshya Adhiniyam, 2023 (effective July 2024). Electronic records presented as secondary evidence require a certificate identifying the device, the method of production, and a responsible signatory.



New Criminal Laws (Effective July 2024)

Bharatiya Nyaya Sanhita (BNS), Bharatiya Nagarik Suraksha Sanhita (BNSS), and Bharatiya Sakshya Adhiniyam (BSA) replaced the IPC, CrPC, and Indian Evidence Act respectively. Digital evidence is now governed under the new legal framework while the IT Act continues to define most cyber-specific offences.



CERT-In Directions, 2022

Specified cyber incidents must be reported to CERT-In within 6 hours; logs to be maintained for 180 days.

Exam-question version: an investigation that is technically perfect but 65B/63-non-compliant produces evidence a court may never see.

TOPIC 3 OF 7

Digital Evidence and Investigative Workflows

03

- What qualifies as digital evidence — and why it is uniquely fragile
- Content versus metadata: the distinction that decides cases
- The six-stage forensic process, and the order of volatility

What Counts as Digital Evidence?

Any information stored or transmitted in digital form that a party may rely on in an investigation or in court.

Four properties that make it unlike physical evidence:



Volatile

RAM contents, network state, and unsaved data vanish on power-off — often permanently.



Easily altered

Opening a file can silently change its timestamps. Handling itself is a threat to integrity.



Duplicable

Perfect copies are possible — which is exactly what forensic imaging exploits, under hash control.



Interpretation-heavy

Bytes rarely speak for themselves; an expert must explain what an artefact means and doesn't mean.

Fragility is why 'preserve first, analyse later' is the iron rule of the field.

Device & System Evidence



Files & remnants

Documents, images, media — plus deleted-file fragments recoverable from unallocated space (the BTK route).



Logs

System, application, and security event logs: who logged in, what ran, what failed, when.



Memory (RAM)

Running processes, decryption keys, chat text never written to disk — captured live or lost forever.



OS artefacts

Browser history, registry entries, recent-file lists, USB connection records, thumbnail caches.



Backups & shadows

Volume shadow copies, phone backups, sync folders — old versions of data users think is gone.



Credential traces

Saved passwords, tokens, Wi-Fi profiles — mapping which accounts and networks the device touched.

Rich, but requires physical or remote access to the device — which is exactly what modern cases often lack.

Network & Online Evidence



Email + full headers

Body content plus routing metadata: servers traversed, originating IP, authentication results. Today's case study.



Messaging

Chats and their metadata — timestamps, delivery status, group membership — from devices or providers.



Social media

Posts, edits, DMs, profile data, engagement trails; obtained by capture or lawful process to platforms.



Traffic & flow data

Packet captures, NetFlow, DNS queries — the network's memory of who connected where.



Cloud & provider records

Login histories, IP logs, file access records held by Google, Meta, Microsoft — reachable via legal process.



Transaction trails

Payment records, UPI trails, crypto ledger entries — following the money remains the oldest technique.

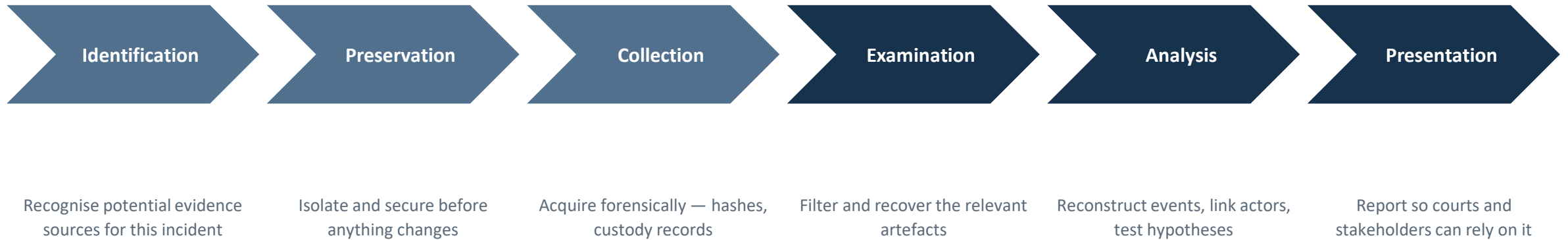
Distributed across providers and borders — powerful, but slower to obtain and jurisdiction-dependent.

Content vs Metadata

ARTEFACT	CONTENT SAYS...	METADATA SAYS...
 Photograph	What the image shows	EXIF: camera model, timestamp, GPS coordinates of capture
 Email	The message text	Headers: true route, originating server, auth results, client used
 Document	The words on the page	Author name, organisation, edit history, total editing time (BTK!)
 Chat message	What was typed	Sent/delivered/read times, device, group membership changes
 Social post	The post itself	Account creation date, login IPs, edit log, amplification pattern

Content tells you what was said. Metadata tells you who, when, where, with what — and whether the content can be trusted.

The Six-Stage Forensic Process



Where AI enters this pipeline

Mostly the darker stages — examination, analysis, presentation: triaging volume, surfacing patterns, drafting summaries. Preservation and custody stay firmly human-controlled.

Memorise the six words — the activity worksheet asks you to map evidence onto them.

Identify, Preserve, Collect: The Groundwork



1 · Identification

- Scope the incident: what happened, when, which systems
- List candidate sources: devices, accounts, logs, providers
- Prioritise: what is most volatile or most likely to vanish?



2 · Preservation

- Isolate devices (airplane mode / Faraday) before they sync or wipe
- Issue legal holds; request providers preserve account data
- Photograph scene and screens; note running state



3 · Collection

- Forensic imaging with write-blockers; live capture where justified
- Hash originals and images; begin custody record
- Capture online content with URLs, IDs, timestamps — not screenshots alone

Mistakes here are unrecoverable — everything downstream analyses only what these stages saved.

Examine, Analyse, Present: Where Meaning Is Made



4 · Examination

- Reduce terabytes to the relevant subset: keyword and hash filtering
- Recover deleted and hidden data; parse artefacts
- AI assist: deduplication, file triage, entity extraction



5 · Analysis

- Reconstruct timelines; correlate across sources
- Link actors, accounts, devices; test alternative explanations
- AI assist: pattern mining, anomaly detection, language analysis

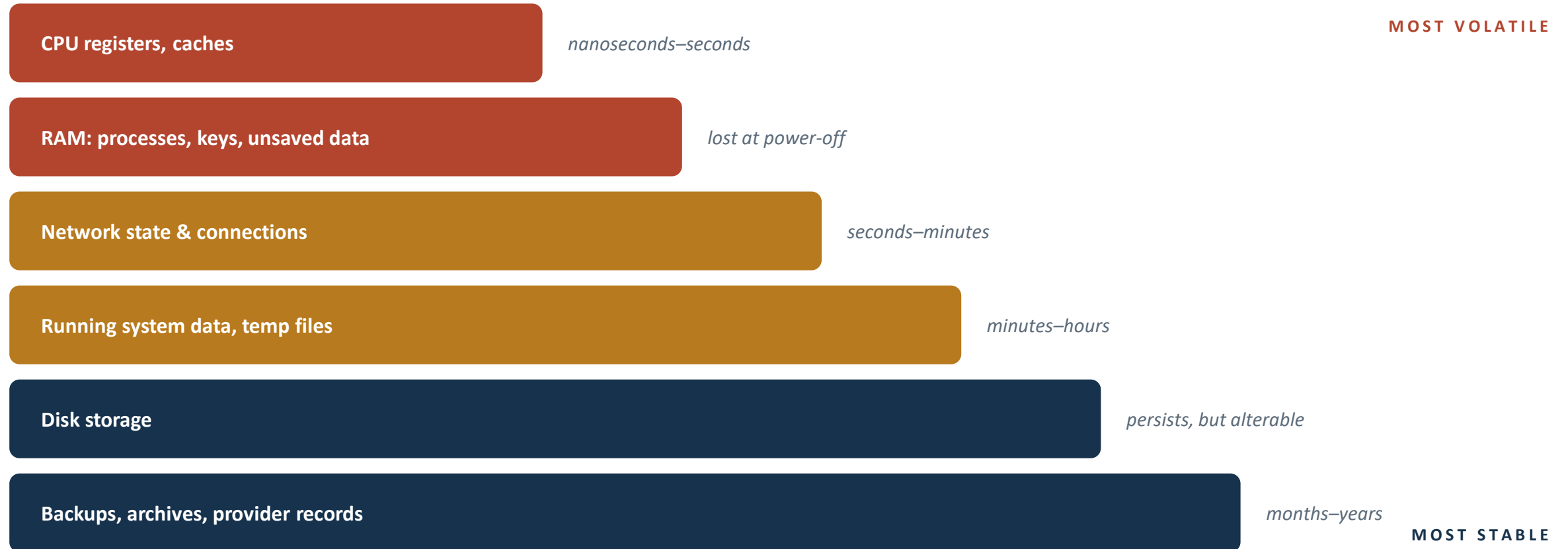


6 · Presentation

- Report findings with methodology, limitations, error sources
- Prepare exhibits and (in India) the s.63 BSA certificate
- AI assist: draft summaries — expert reviews and owns every word

Notice the pattern: AI assists inside each stage; a named human remains answerable for each stage.

Order of Volatility: Collect the Fastest-Dying First



Codified in RFC 3227: when everything cannot be captured, capture in this order — top first.

TOPIC 4 OF 7

Role of Artificial Intelligence in Cybersecurity

04

- Why defence outgrew human reading speed
- Sorting the vocabulary: AI, ML, deep learning, generative AI
- Six real functions AI performs today — and four hard limits

Why Defence Outgrew Human Reading Speed



Alert flood

A mid-size SOC routinely faces thousands of alerts per day; most are noise, a handful are incidents. Humans fatigue; models don't.



Attacker speed

Automated attacks move in seconds — credential stuffing, worm propagation. WannaCry (2017) hit organisations in 150+ countries within days.



Analyst scarcity

The ≈4M global workforce gap means most organisations will never hire their way out. Force-multiplication is the only path.

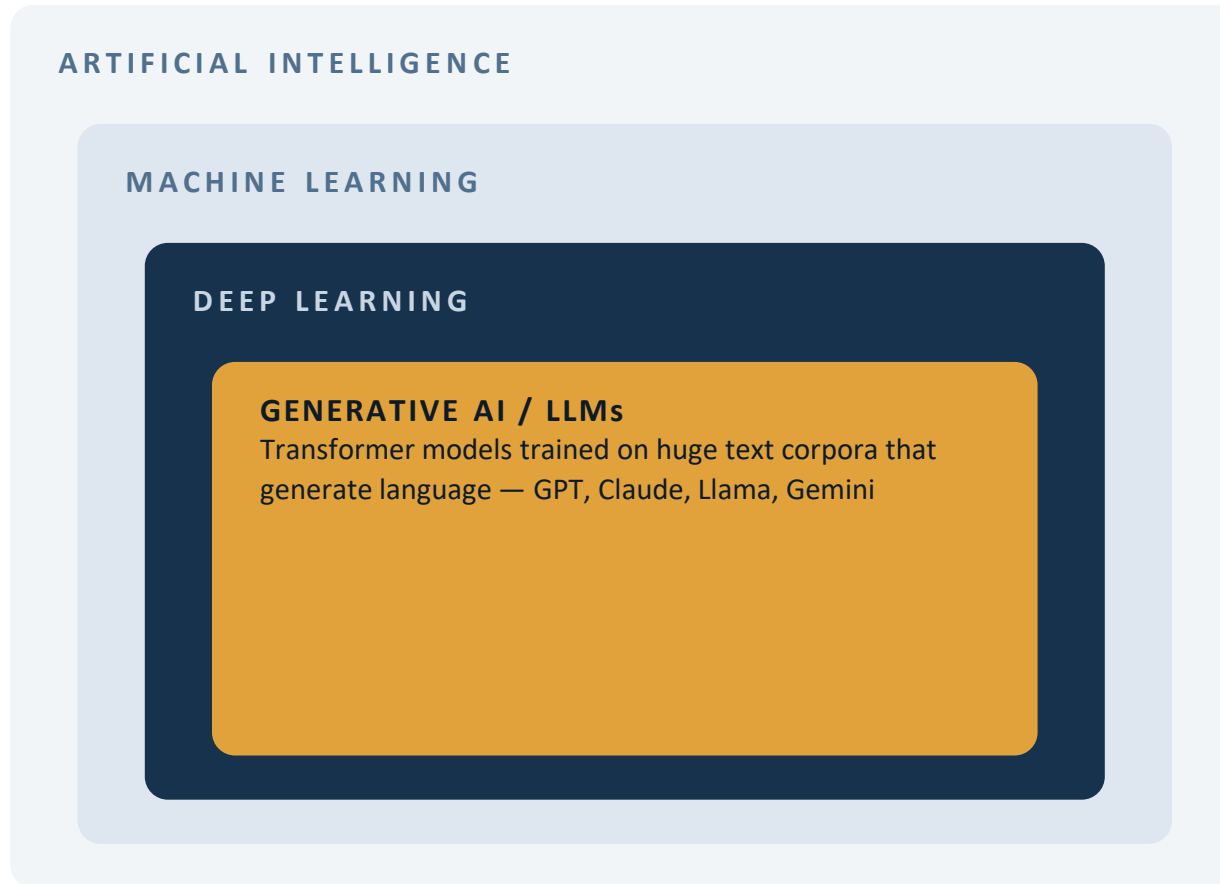


Data gravity

Logs, packets, emails, posts — evidence is text and telemetry at petabyte scale. Reading is now a machine task; judging remains a human one.

AI in security is not a luxury upgrade — it is the only response to an arithmetic problem.

AI, ML, Deep Learning, GenAI — Sorted



AI — Any technique making machines perform tasks that need intelligence — includes 1980s rule-based expert systems.

ML — Systems that learn patterns from data instead of following hand-written rules. Spam filters live here.

Deep learning — ML using multi-layer neural networks; powers image, speech, and modern language models.

GenAI / LLMs — Deep models that produce new content. The engine of Sessions 3–4 — as tool and as threat.

Six Things AI Actually Does in Cyber Defence



Anomaly & intrusion detection

Learn normal network and user behaviour; flag deviations rule-based systems miss.



Malware & phishing classification

Separate malicious binaries, URLs, and emails from benign traffic at machine speed.



Alert triage & prioritisation

Rank thousands of daily SIEM alerts so analysts spend time on the few that matter.



Threat intelligence mining

NLP reads advisories, feeds, forums — extracting indicators, actors, and tactics.



Behaviour analytics (UEBA)

Profile accounts over time to expose insider misuse and stolen credentials.



Evidence summarisation

LLMs condense huge disclosure sets and logs into reviewable investigative leads.

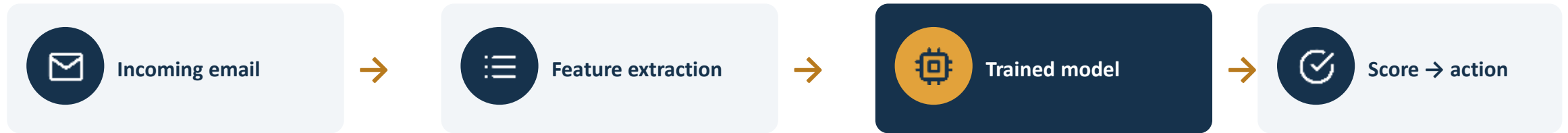
Common thread: AI compresses analyst attention — it does not replace analyst judgement.

AI Across the Six Forensic Stages

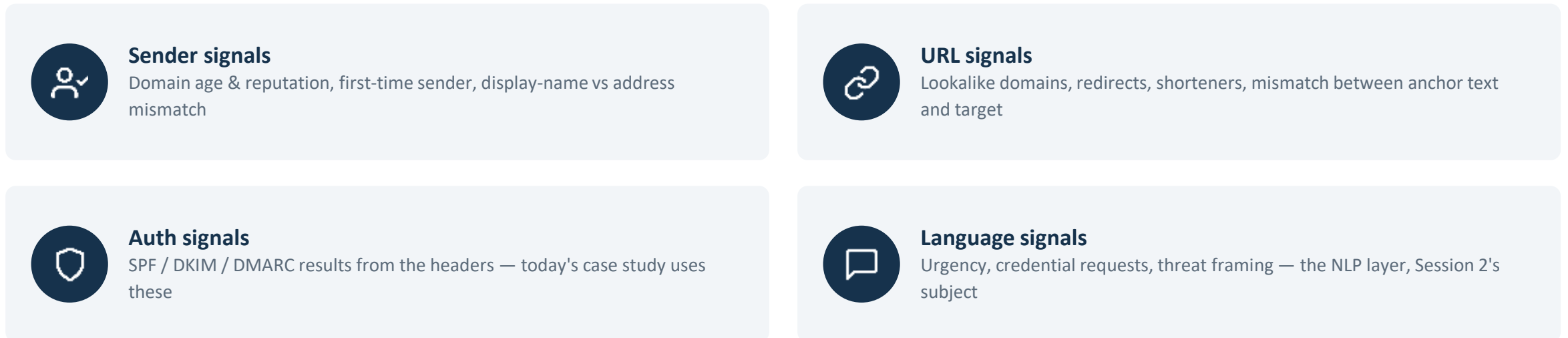
STAGE		AI CONTRIBUTION	HUMAN KEEPS
	Identification	Entity and link discovery across sources; scoping suggestions	Scope decision
	Preserve + Collect	Minimal — flagging what to prioritise for capture	Every action
	Examination	Deduplication, file triage, keyword expansion, NER	Relevance calls
	Analysis	Timeline reconstruction, anomaly detection, language analysis	Interpretation
	Presentation	First-draft summaries and visualisations	Final report & oath

The right-hand column is the exam answer to 'will AI replace forensic examiners?' — no; it re-allocates their hours.

Inside an ML Phishing Filter



What the model looks at (typical feature families):



Every mail provider runs some version of this on billions of messages daily — invisible, until it misses.

Four Hard Limits of AI in Security



False positives at scale

99% accuracy on a million daily events still means thousands of wrong flags. Precision failures create their own workload — and desensitise analysts.



Adversarial evasion

Attackers probe and adapt to detectors: tweaked malware, reworded phishing. Every deployed model becomes a target.



Bias & data debt

Models inherit the gaps and skews of training data — a detector trained on English scams misses Hinglish ones. Session 6's core theme.



No accountability

A model cannot take an oath, be cross-examined, or bear responsibility. Legally and ethically, a human must own every conclusion.

The honest posture: AI as instrument, examiner as author. Anything stronger is over-claiming.

TOPIC 5 OF 7

Introduction to Language-Based Evidence

05

- Why most cybercrime leaves a linguistic trail
- Five things language reveals — authorship, intent, deception, coordination, provenance
- Anatomy of an email header: the skill the case study needs

Text Is Evidence

Most cybercrime leaves a linguistic trail: phishing emails, threatening messages, fraudulent posts, ransom notes, scam scripts. The words themselves — and how they are written — carry investigative signal.



What language can reveal



Authorship Stylometric habits — vocabulary, punctuation, error patterns — can link texts to a likely writer.



Intent & threat level Distinguishing venting from planning; grading urgency of threatening communication.



Deception cues Hedging, distancing language, and inconsistency patterns associated with deceptive text.



Networks & coordination Shared phrasing and timing across accounts exposing coordinated campaigns.



Provenance Increasingly: was this written by a human at all, or generated by a model?

Session 2 turns each of these into hands-on NLP practice.

The Unabomber: Convicted by His Own Prose

1978–1995

A serial bomber kills 3 and injures 23 across the US. Seventeen years of physical forensics yield no suspect.

1995

He demands newspapers publish his 35,000-word manifesto, 'Industrial Society and Its Future'. The FBI agrees.

The break

David Kaczynski recognises his brother Ted's ideas and phrasing — including the inverted proverb 'you can't eat your cake and have it too'.

The method

FBI linguistic analysis matches the manifesto's idiolect to Ted's private letters — supporting the search warrant.

1996–98

Kaczynski is arrested at his Montana cabin; the typewriter and drafts inside seal the case.



Why it matters here

The case that legitimised forensic linguistics: writing style treated as identifying evidence, defensible in court.

Modern stylometry automates this comparison — NLP models measuring what one FBI analyst once did by hand.

BTK fell to metadata; Kaczynski fell to style. Both are language-adjacent evidence — Topic 5's whole point.

Anatomy of an Email Header

```
1 Return-Path: <bounce@mail-relay77.example-esp.com>

2 Received: from unknown (HELO srv12) [185.220.xx.xx]

    by mx.recipient.com; Mon, 20 Jul 2026 14:02:11 +0530

3 Authentication-Results: mx.recipient.com;

    spf=fail dkim=none dmarc=fail

4 Message-ID: <a91x@mail-relay77.example-esp.com>

5 From: "Anil Mehta, CFO" <a.mehta@rneridian-industries.com>

6 Reply-To: anil.mehta.cfo.office@gmail.com

Subject: URGENT - Confidential wire transfer
```

1 • Return-Path — Where bounces go — often a bulk-mail relay, not the claimed company.

2 • Received chain — Servers traversed, read bottom-up. The first hop carries the true origin.

3 • Auth results — SPF / DKIM / DMARC verdicts. fail = sending server not authorised for that domain.

4 • Message-ID — Its domain should match the sender's organisation. Here it doesn't.

5 • From — Freely forgeable display text. Note the lookalike: rneridian vs meridian.

6 • Reply-To — Where replies actually go — a free webmail address. Classic BEC tell.

Ten minutes from now you will read a full header yourselves — this is the decoding key.

Human or Machine? The Provenance Problem



Why it now matters forensically

- Stylometry assumes a human idiolect; LLM text has no stable personal style to match
- A suspect can claim 'an AI wrote that' — prosecutors must be able to respond
- Scam scripts, fake reviews, and influence posts are increasingly machine-drafted



State of detection

- AI-text detectors exist but are unreliable: high false-positive rates, easily defeated by light editing
- No detector output should ever be treated as proof — corroborate with metadata and context
- Research frontier: watermarking and provenance standards (C2PA) — Session 4 continues this

Rule for practice: treat 'AI-generated?' as a hypothesis to investigate, never as a verdict a tool can hand you.

TOPIC 6 OF 7

Challenges in AI-Driven Cyber Threats

06

- The adversary's AI toolkit — six capabilities, all live today
- Two documented cases: a US\$25M deepfake call, a wartime deepfake broadcast
- What this breaks for investigators, and the admissibility question

The Adversary Has AI Too



AI-generated phishing

Fluent, personalised lures at scale — the tell-tale bad grammar of old scams is gone.



Deepfakes & voice cloning

Synthetic video and audio for fraud, impersonation, and executive-level social engineering.



Synthetic identities & bots

Model-generated personas and profile content powering influence and fraud networks.



Automated attack tooling

LLM-assisted reconnaissance, malware variation, adaptive attack scripts — WormGPT-style services sold openly.



Misinformation at scale

Generative models producing coordinated false narratives faster than manual fact-checking.



Prompt injection & model abuse

Attacks aimed at the AI systems organisations now depend on — a brand-new attack surface.

Every capability that assists an investigator can be repurposed by an attacker — often first.

The US\$25 Million Video Call That Wasn't

Jan 2024

A finance employee at engineering firm Arup's Hong Kong office receives an email about a 'confidential transaction' from the UK CFO.

Suspicion

It smells like phishing. He hesitates — correctly.

The call

He joins a video conference. The CFO is there. Colleagues he recognises are there. Everyone looks and sounds right.

The catch

Every other participant is an AI-generated deepfake, built from publicly available footage.

The loss

Reassured, he makes 15 transfers totalling ≈ HK\$200M (~US\$25.6M) to mule accounts.



Forensic lessons

- His phishing instinct was right — the deepfake defeated his verification step, not his suspicion
- 'I saw them on video' is no longer authentication
- The investigable trail was everywhere else: the email headers, the call platform logs, the mule accounts

Reported by Hong Kong police, February 2024 — widely covered; Arup later confirmed being the target.

Three More Documented Incidents



Zelensky deepfake (2022)

Days into the Ukraine invasion, a deepfake video of President Zelensky telling soldiers to surrender spread online and appeared on a hacked news site. Debunked within hours — but a proof of concept for wartime synthetic media.



WormGPT / FraudGPT (2023)

Subscription LLM tools marketed on underground forums for writing BEC lures and malicious code — no ethics filters, pay in crypto. Generative AI as a criminal service, documented by multiple security vendors.



Voice-clone scams (ongoing)

Cloned voices of relatives and executives used in distress and payment scams worldwide, including cases reported across India. Seconds of public audio are enough to build a working clone.

Session 4 dissects the deepfake cases; Session 3 dissects the GenAI-phishing pipeline.

What AI-Driven Threats Break for Investigators

CHALLENGE

WHY IT MATTERS FOR FORENSICS



Human vs machine text

Authorship and stylometric methods assume a human writer; generated text weakens that assumption.



Attribution

AI intermediaries blur the link between content and its true originator across accounts and borders.



Speed & scale

Synthetic content is produced faster than manual verification can keep up with.



Seeing is no longer believing

Audio and video — once the strongest evidence classes — now require their own authentication.



Admissibility

Legal standards for AI-assisted findings are still forming — validation and error rates matter.

Will a Court Accept AI-Assisted Findings?

Questions any tribunal — Indian or otherwise — will eventually ask of an AI-assisted forensic conclusion:



Is the method validated?

Known error rates, tested on relevant data, documented limitations — not a vendor's marketing accuracy.



Can it be explained?

A black-box 'fake: 0.93' survives no cross-examination. Explainable outputs — Sessions 5–6 — exist for exactly this reason.



Who is responsible?

A qualified human expert must adopt, verify, and testify to every AI-assisted finding as their own opinion.



Is the trail intact?

Versions, prompts, parameters, and data lineage logged — the chain of custody now extends into the model pipeline.

This is why 'Explainable AI' is a full day of this series, not a buzzword — explainability is an admissibility strategy.

TOPIC 7 OF 7

Emerging Trends in Intelligent Forensic Systems

07

- Four directions the field is moving — mapped to Sessions 2–10
- What LLM-assisted investigation concretely looks like
- A first glimpse of explainable AI, and open research problems

Four Directions, Nine Sessions



LLM-assisted investigation

Models that read, summarise, and question evidence alongside the analyst — humans validating every finding.

→ Sessions 3–4



Explainable AI (XAI)

SHAP, LIME, and attention-based methods turning model outputs into court-defensible reasoning.

→ Sessions 5–6



Social media forensics

Detecting rumours, abuse, and coordinated manipulation on the platforms where harm occurs.

→ Sessions 7–8



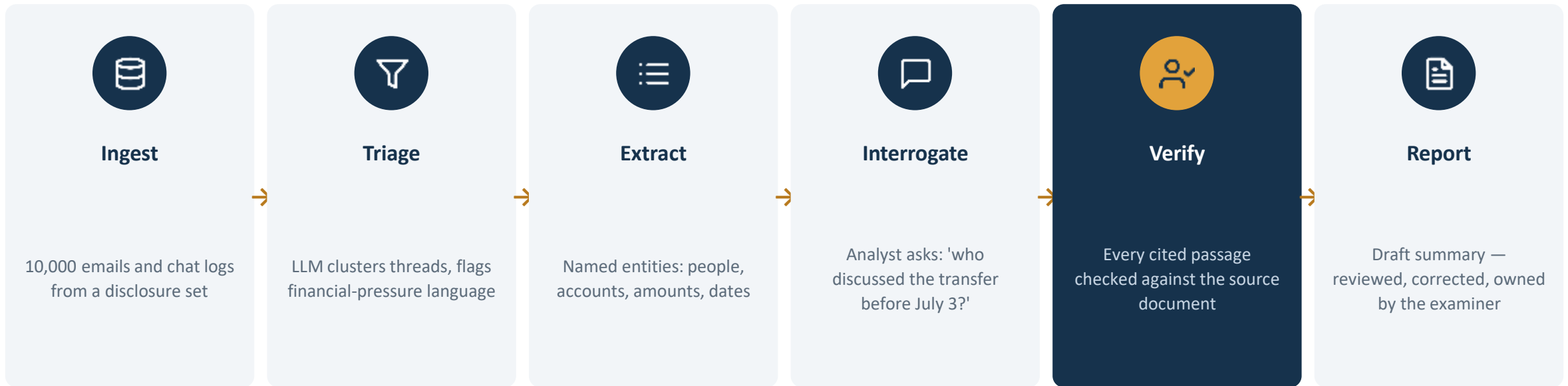
Multimodal & agentic systems

Joint analysis of text, images, audio, logs; AI agents automating investigative sub-tasks.

→ Sessions 9–10

Today built the vocabulary; the series walks each road with hands-on tools.

What 'LLM-Assisted Investigation' Looks Like



The dark box is the whole game: LLMs hallucinate. Verification against sources is what separates an investigative tool from a liability.

A First Glimpse of Explainable AI

The same phishing verdict, two ways:

BLACK BOX

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"phishing_probability = 0.94"
```

Accurate, perhaps. Defensible under cross-examination? No — there is nothing to examine.

EXPLAINED (illustrative SHAP-style view)

SPF/DMARC failure



Reply-To ≠ From domain



Urgency language



First-time sender



Payment request



Each factor's contribution is visible — an expert can check, weigh, and defend it.

Day 3 builds these views yourselves with SHAP and LIME in Colab.

Open Problems Worth a Paper

-  **Stylometry after LLMs**

Can authorship attribution be rebuilt for a world where suspects can machine-laundry their style?
-  **Code-mixed & low-resource forensics**

Hinglish and Indian-language scam text evades English-centric models — a locally urgent, globally underserved problem.
-  **Explainability that courts accept**

What form of model explanation actually satisfies legal scrutiny? An open HCI + law + ML question.
-  **Coordination detection**

Distinguishing organic virality from orchestrated campaigns, with defensible error rates.
-  **Benchmarks & validation**

The field lacks shared, realistic evaluation datasets for AI-forensic tools — building one is a contribution.

Session 10 runs a research-problem formulation workshop — start noting your candidates now.

CASE STUDY · 20 MINUTES

Investigating Suspicious Email Communications

- A curated incident inspired by real-world BEC (Business Email Compromise) cases — fully fictional entities
- You will read the actual email and its headers, and find the red flags
- Then: the Digital Evidence Identification Exercise (worksheet, 20 minutes)



SESSION 1 · CASE STUDY · 20-MINUTE GUIDED LAB

SCAN ME



ANSWER KEY

Incident Brief: Meridian Industries

Monday, 20 July, 14:02. Priya Nair, finance manager at Meridian Industries (a fictional Pune manufacturer), receives an email that appears to come from the company's CFO, Anil Mehta — marked urgent and confidential, requesting an immediate wire transfer of ₹48,00,000 to a 'new vendor' before 3 PM. Mr Mehta is travelling and unreachable by phone. Priya hesitates and forwards the email to IT security. You are the investigator.



Why the timing matters

3 PM deadline + unreachable executive = engineered pressure. BEC scripts manufacture exactly this squeeze.



What Priya did right

She paused and escalated instead of replying. The evidence chain starts intact because of that choice.



Your task

Examine the email as evidence: headers first, language second, then map next investigative steps.

The Email, As Received

Inbox — URGENT - Confidential wire transfer

From: "Anil Mehta, CFO" <a.mehta@rneridian-industries.com>

Reply-To: anil.mehta.cfo.office@gmail.com

To: priya.nair@meridian-industries.com

Date: Mon, 20 Jul 2026 14:02:11 +0530

Priya,

I am in meetings with the acquisition partners and cannot take calls. We must complete a confidential payment of Rs 48,00,000 to our new vendor Apex Component Traders before 3 PM today, or we lose the contract penalty clause.

This is highly sensitive — do not discuss with anyone in the office until the announcement. Process it under 'miscellaneous procurement' and confirm to me on this email only. I am counting on you to handle this personally.

Sent from my iPhone
Anil

Read it as Priya would — 90 seconds. Note anything that feels wrong before we dissect it together.

Technical Red Flags (Headers & Addresses)

-  **Lookalike domain**

rneridian-industries.com — the letters 'r'+ 'n' imitate 'm'. Registered days ago; the real domain is meridian-industries.com.
-  **Reply-To mismatch**

Replies silently divert to a free Gmail address the attacker controls — the classic BEC redirection.
-  **Authentication failures**

SPF fail, DKIM none, DMARC fail: the sending server was never authorised to send for that domain.
-  **Route anomalies**

Received chain shows a bulk-mail relay and an origin IP unrelated to the company's mail infrastructure.
-  **Just-in-time registration**

WHOIS-style checks on the lookalike domain reveal creation date within the past week — a near-certain fraud signal.

None of these are visible in the email body — which is exactly why headers are examined first.

Linguistic & Behavioural Red Flags



Manufactured urgency

“before 3 PM today” + penalty threat — compressing the victim's time to think or verify.



Authority invocation

CFO identity plus “I am counting on you personally” — exploiting hierarchy and flattery together.



Secrecy instruction

“do not discuss with anyone” — isolating the victim from the colleagues who would catch the fraud.



Channel restriction

“confirm to me on this email only” + conveniently unreachable by phone — blocking out-of-band verification.



Procedural bypass

“process under miscellaneous procurement” — explicitly instructing the victim to defeat internal controls.

Individually, each line could be innocent. The cluster is the signature — and clusters are what NLP models score.

Business Email Compromise (BEC)

BEC is a cyberattack in which attackers use fraudulent or spoofed emails to impersonate trusted individuals and manipulate victims into making financial transactions or disclosing confidential information.

How a BEC Attack Works

- The attacker researches the organisation.
- They impersonate a senior executive, supplier, or partner.
- They send a convincing email requesting an urgent payment or sensitive information.
- The victim believes the request is genuine and acts on it.
- The money is transferred to the attacker's account.

Common Characteristics

- Urgent payment request
- CEO/CFO impersonation
- Look-alike domain (e.g., **rneridian.com** instead of **meridian.com**)
- Reply-To address different from the From address
- Requests for secrecy
- Pressure to bypass normal approval procedures
- New or changed bank account details

Business Email Compromise (BEC)

Real-World Example

In your case study:

The attacker impersonates **Anil Mehta (CFO)**.

Priya receives an urgent request to transfer **₹48,00,000**.

The CFO is "travelling" and "cannot take calls."

The sender uses a **look-alike domain** and a different **Reply-To** address.

The email creates urgency and instructs Priya not to discuss it with anyone.

These are classic indicators of a **Business Email Compromise (BEC)** attack.

BEC is a social engineering attack that relies on trust and deception rather than malware.

The attack succeeds when victims believe they are communicating with a trusted person and act without independent verification.

Questions Worth Arguing About

Q1

Priya forwarded the email to IT. Did forwarding alter or endanger any evidence? What would have been ideal handling?

Q2

The lookalike domain was registered abroad and paid for in crypto. Realistically, how far can attribution go — and through which legal channels?

Q3

If an AI triage model had quarantined this email automatically, what would the organisation have lost, if anything?

Q4

Suppose the attacker had used an LLM to write a flawless, personalised message with none of our five language cues. Which red flags survive?

Q5

For the researchers: which part of this case would make the best benchmark task for an AI-forensics dataset?

WRAP-UP

Key Takeaways

1

Digital forensics evolved from device recovery to intelligent, large-scale evidence interpretation — but its five principles are unchanged.

2

Metadata is the quiet decider: BTK's document properties, Kaczynski's idiolect, our email's headers.

3

AI compresses analyst attention across triage, detection, and summarisation — under human judgement, with explainability as the admissibility strategy.

4

The adversary industrialised first: GenAI phishing and deepfakes mean verification procedures, not perception, are the last line.

**UP NEXT · SESSION 2****NLP for Cybersecurity & Forensic Analytics**

Hands-on: entity extraction, suspicious-keyword analysis, and a first taste of authorship attribution — spaCy, NLTK, Colab.

Questions & discussion